C. The Legend of Freya the Frog

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: implementation, math | https://codeforces.com/problemset/problem/2009/C

Freya the Frog is traveling on the 2D coordinate plane. She is currently at point (0,0) and wants to go to point (x,y). In one move, she chooses an integer d such that $0 \le d \le k$ and jumps d spots forward in the direction she is facing.

Initially, she is facing the positive x direction. After every move, she will alternate between facing the positive x direction and the positive y direction (i.e., she will face the positive y direction on her second move, the positive x direction on her third move, and so on).

What is the minimum amount of moves she must perform to land on point (x,y)?

Input

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

Each test case contains three integers x, y, and k ($0 \le x, y \le 10^9$, $1 \le k \le 10^9$).

Output

For each test case, output the number of jumps Freya needs to make on a new line.

Example

Input
3 9 11 3 0 10 8 1000000 100000 10

Output
8 4 199999

Note

In the first sample, one optimal set of moves is if Freya jumps in the following way: (0,0) → (2,0) → (2,2) → (3,2) → (3,5) → (6,5) → (6,8) → (9,8) → (9,11). This takes 8 jumps.